

Menu-Driven Integer Tables in C

Nested loops, switch-case, input validation, compilation and testing

Aim

To write a menu-driven C program that accepts two positive integers m and n ($m \leq n$) and repeatedly performs multiplication, addition, subtraction, or division-table operations until the user selects Exit.

Theory / How the code works

A do-while loop validates m and n and another do-while loop keeps displaying the menu until choice 5 is selected. A switch statement selects the required operation. For choices 1 to 4, nested for loops are used: the outer loop moves from m to n , while the inner loop uses values 1 to 10. Division uses type casting to double and `%.2f` so the answer is displayed to two decimal places.

Algorithm

1. Start.
2. Read m and n .
3. If m or n is non-positive, or $m > n$, display an error and read them again.
4. Display the menu: 1-Multiples, 2-Addition, 3-Subtraction, 4-Division, 5-Exit.
5. Read the user choice.
6. Use `switch(choice)` to select the operation.
7. For choice 1, for every integer i from m to n , print $i \times j$ for $j = 1$ to 10.
8. For choice 2, for every integer i from m to n , print $i + j$ for $j = 1$ to 10.
9. For choice 3, for every integer i from m to n , print $i - j$ for $j = 1$ to 10.
10. For choice 4, for every integer i from m to n , print i / j for $j = 1$ to 10 using floating-point division and two decimal places.
11. For choice 5, exit the loop. For any other choice, display an invalid-choice message.
12. Repeat the menu until choice 5 is selected. Stop.

Test-Case Table

Case	Input	Choice	Expected result
1	$m=2, n=3$	1	Print first 10 multiples of 2 and 3
2	$m=2, n=3$	2	Print addition tables of 2 and 3, addends 1-10
3	$m=2, n=3$	3	Print subtraction tables of 2 and 3, subtrahends 1-10
4	$m=2, n=3$	4	Print division results for 2 and 3 by 1-10 to 2 decimals
5	$m=5, n=5$	1	Boundary: print multiples for the single integer 5
6	$m=0, n=4$ then 2,4	-	Reject invalid pair, then accept 2,4
7	$m=6, n=3$ then 3,6	-	Reject $m>n$, then accept valid pair
8	$m=2, n=3$	9	Display invalid menu choice and show menu again

Cas e	Input	Choice	Expected result
9	m=2, n=3	5	Exit program

C Program

```
#include <stdio.h>

int main(void) {
    int m, n, choice, i, j;

    do {
        printf("Enter two positive integers m and n (m <= n): ");
        scanf("%d %d", &m, &n);
        if (m <= 0 || n <= 0 || m > n)
            printf("Invalid input! Enter positive integers with m <= n.\n");
    } while (m <= 0 || n <= 0 || m > n);

    do {
        printf("\nMENU\n");
        printf("1. First 10 multiples\n");
        printf("2. Addition tables\n");
        printf("3. Subtraction tables\n");
        printf("4. Division tables\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                for (i = m; i <= n; i++) {
                    printf("\nMultiples of %d:\n", i);
                    for (j = 1; j <= 10; j++)
                        printf("%d x %d = %d\n", i, j, i * j);
                }
                break;

            case 2:
                for (i = m; i <= n; i++) {
                    printf("\nAddition table of %d:\n", i);
                    for (j = 1; j <= 10; j++)
                        printf("%d + %d = %d\n", i, j, i + j);
                }
                break;

            case 3:
                for (i = m; i <= n; i++) {
                    printf("\nSubtraction table of %d:\n", i);
                    for (j = 1; j <= 10; j++)
                        printf("%d - %d = %d\n", i, j, i - j);
                }
                break;

            case 4:
                for (i = m; i <= n; i++) {
                    printf("\nDivision table of %d:\n", i);
                    for (j = 1; j <= 10; j++)
                        printf("%d / %d = %.2f\n", i, j, (double)i / j);
                }
                break;

            case 5:
                printf("Exiting program.\n");
                break;

            default:
                printf("Invalid menu choice! Please enter 1 to 5.\n");
        }
    } while (choice != 5);

    return 0;
}
```

Compilation and Execution

The program was compiled with GCC using **gcc -Wall -Wextra -pedantic menu_tables.c -o menu_tables**. Compilation completed successfully. It was then executed with $m = 2$, $n = 3$ and choices 1, 2, 3, 4, an invalid choice 9, and finally 5.

Sample Output

```
Enter two positive integers m and n (m <= n):
MENU
1. First 10 multiples
2. Addition tables
3. Subtraction tables
4. Division tables
5. Exit
Enter your choice:
Multiples of 2:
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

Multiples of 3:
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12

... (other selected tables printed) ...

1. First 10 multiples
2. Addition tables
3. Subtraction tables
4. Division tables
5. Exit
Enter your choice:
Division table of 2:
2 / 1 = 2.00
2 / 2 = 1.00
2 / 3 = 0.67
2 / 4 = 0.50
2 / 5 = 0.40
2 / 6 = 0.33
2 / 7 = 0.29
2 / 8 = 0.25
2 / 9 = 0.22
2 / 10 = 0.20

Division table of 3:
3 / 1 = 3.00
3 / 2 = 1.50
3 / 3 = 1.00
3 / 4 = 0.75
3 / 5 = 0.60
3 / 6 = 0.50
3 / 7 = 0.43
3 / 8 = 0.38
3 / 9 = 0.33
3 / 10 = 0.30

MENU
1. First 10 multiples
2. Addition tables
3. Subtraction tables
4. Division tables
5. Exit
Enter your choice: Invalid menu choice! Please enter 1 to 5.

MENU
1. First 10 multiples
2. Addition tables
3. Subtraction tables
4. Division tables
5. Exit
```

Enter your choice: Exiting program.

Result

The program successfully performs all four table operations using switch-case and nested loops, validates the range $m \leq n$ with positive integers, reports invalid menu choices, displays division answers to two decimal places, and repeats until Exit is selected.